

AWS DEVOPS INTERVIEW GUIDE

Scenario 1: Design a Disaster Recovery Solution for a Production Web Application

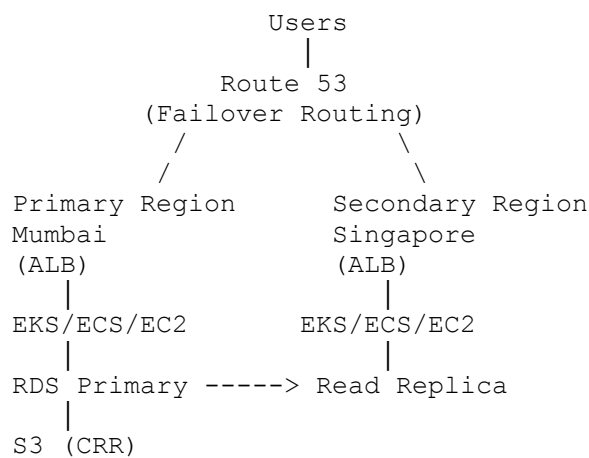
Scenario

Your company's production application is hosted in the **Mumbai (ap-south-1)** region. Management requires that if the Mumbai region becomes unavailable, users should automatically be redirected to another AWS region within **15 minutes**, with minimal data loss.

How would you design the DR solution?

Solution

Architecture



Components

- Route 53 Failover Routing
 - ALB in both regions
 - EKS/ECS/EC2 deployed in both regions
 - Cross-Region S3 Replication (CRR)
 - RDS Cross-Region Read Replica
 - CloudWatch Health Checks
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Failover Process

1. Mumbai becomes unavailable.
 2. Route 53 health check fails.
 3. DNS switches to the Singapore ALB.
 4. Promote the Read Replica to a standalone database (if required).
 5. Users continue accessing the application from Singapore.
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Interview Points

- RTO: ~15 minutes
 - RPO: Seconds to minutes (depends on replication lag)
 - Automatic DNS failover
 - Multi-Region architecture
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Scenario 2: Cross-Region S3 Disaster Recovery

Scenario

Your production application stores customer documents in an S3 bucket in Mumbai. If the region fails, documents must still be available from another region.

How would you implement this?

Solution

Use **S3 Cross-Region Replication (CRR)**.

Architecture

Users

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S3 Bucket (Mumbai)

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Cross Region Replication

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S3 Bucket (Singapore)

Steps

1. Create destination bucket.
2. Enable Versioning on both buckets.
3. Configure CRR.
4. Create an IAM role for replication.
5. Verify replication status.

Benefits

- Automatic replication
- Durable backups
- Low RPO
- Supports versioned objects

Interview Tip

CRR replicates **new objects only**. Existing objects require **S3 Batch Replication** or a one-time copy operation.

Scenario 3: Disaster Recovery for Amazon RDS

Scenario

Your PostgreSQL production database is hosted in Mumbai. The application must continue running even if the Mumbai region becomes unavailable.

How would you design the DR solution?

Solution

Architecture

Mumbai

RDS Primary

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Cross Region Read Replica

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Singapore

During Disaster

Mumbai Down

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Promote Read Replica

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Application connects to Singapore

Recovery Steps

1. Create a Cross-Region Read Replica.
2. Monitor replication lag.
3. During a regional failure:
 - Promote the replica.
 - Update application configuration or Route 53.
4. Resume application traffic.

RTO

Approximately 10–20 minutes.

RPO

Typically seconds to minutes, depending on replication lag.

Scenario 4: Cross-Region Disaster Recovery for EKS

Scenario

Your production Kubernetes cluster runs in Mumbai. If the region fails, services must be restored quickly in another region.

How would you implement this?

Solution

Architecture

EKS Mumbai

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Container Images (ECR Replication)

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EKS Singapore

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Route53 Failover

Requirements

- EKS clusters in both regions
- Infrastructure defined using Terraform or CloudFormation
- Container images replicated (Amazon ECR replication)
- Secrets synchronized securely
- Persistent storage replicated or restored
- Route 53 failover

Recovery Process

1. Mumbai fails.
2. Route 53 redirects traffic.
3. Applications start in Singapore.
4. Database is promoted.
5. Users continue using the application.

Scenario 5: Terraform-Based Disaster Recovery

Scenario

The company wants the entire AWS infrastructure to be recreated automatically in another region.

How would you design this?

Solution

Use **Terraform**.

Components

Terraform

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VPC

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EKS

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RDS

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ALB

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Route53

Best Practices

- Use reusable modules.
- Parameterize regions using variables.
- Store Terraform state remotely (for example, Amazon S3 with state locking).
- Validate in a staging region before production.

Advantages

- Infrastructure as Code
- Fast provisioning
- Consistent deployments
- Version control

Scenario 6: Cross-Region CI/CD Pipeline

Scenario

Your CI/CD pipeline deploys only to Mumbai. During a regional outage, deployments must continue to Singapore.

How would you achieve this?

Solution

Architecture

GitHub

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Jenkins

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Build Docker Image

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Amazon ECR

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Deploy

├─ Mumbai
└─ Singapore

Steps

1. Push image to ECR.
2. Replicate image to another region.
3. Deploy to both EKS/ECS clusters.
4. Use Route 53 Failover Routing.

Benefits

- Continuous deployment during outages
- Faster disaster recovery
- Consistent application versions

Scenario 7: Disaster Recovery Testing

Scenario

Management wants proof that the Disaster Recovery solution actually works.

How would you perform a DR drill?

Solution

Steps

1. Schedule a maintenance window.
2. Back up production data.
3. Simulate a regional outage.
4. Verify:
 - Route 53 failover
 - Database promotion
 - Application health
 - Monitoring and alerts
5. Record:
 - Actual RTO
 - Actual RPO
6. Restore the original environment.

Deliverables

- Test report
- Recovery timeline
- Issues found
- Improvements identified

Scenario 8: Cross-Region Secrets Management

Scenario

Applications in both regions require the same database credentials and API keys.

How would you manage secrets?

Solution

Use **AWS Secrets Manager**.

Architecture

Secrets Manager

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Primary Region

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Replicate Secret

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Secondary Region

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Applications

Benefits

- Consistent credentials
- Centralized management
- Automatic rotation support
- Reduced manual effort

Scenario 9: Monitoring During Disaster Recovery

Scenario

During a regional outage, the operations team needs immediate visibility into failures and recovery.

How would you monitor the environment?

Solution

Use:

- Amazon CloudWatch
- CloudWatch Alarms
- Amazon SNS
- AWS CloudTrail
- AWS Health Dashboard

Monitor

- ALB health
- EC2/ECS/EKS status
- RDS replication lag
- Lambda errors

- Route 53 health checks

Notifications

CloudWatch Alarm

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SNS

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Email / SMS / Slack

Scenario 10: Which Disaster Recovery Strategy Would You Choose?

Scenario

Your company asks you to recommend the appropriate DR strategy based on business requirements.

How would you decide?

Solution

Strategy	Description	Cost	RTO	RPO	Best Use Case
Backup & Restore	Recover from backups	Lowest	Hours	Hours	Non-critical applications
Pilot Light	Core services always running	Low	Tens of minutes	Minutes	Medium-critical workloads
Warm Standby	Smaller but functional environment	Medium	Minutes	Minutes	Business-critical applications
Multi-Site Active/Active	Full production in multiple regions	Highest	Near Zero	Near Zero	Mission-critical applications

Interview Tip

Always choose the strategy based on:

- Business requirements
- Budget
- Compliance needs
- Required RTO

- Required RPO

For example:

- **Internal HR application** → Backup & Restore
- **Banking system** → Multi-Site Active/Active
- **E-commerce platform** → Warm Standby or Pilot Light, depending on availability requirements

Cross-Region Disaster Recovery Interview Cheat Sheet

AWS Service	Disaster Recovery Role
Route 53	DNS failover and traffic routing
Amazon S3	Cross-Region Replication (CRR) and backup storage
Amazon RDS	Cross-Region Read Replicas for database recovery
Amazon ECR	Cross-region image replication
Amazon EKS/ECS	Multi-region application deployment
Elastic Load Balancer	Regional traffic distribution
AWS Lambda	Serverless processing in both regions
AWS Secrets Manager	Replicate secrets across regions
Amazon CloudWatch	Monitoring and alarms
Amazon SNS	Disaster notifications
AWS Backup	Centralized backup management
Terraform/CloudFormation Infrastructure recreation and automation	

Production Best Practices

- Define **RTO** and **RPO** before designing the DR architecture.
- Automate infrastructure provisioning with **Terraform** or **CloudFormation**.
- Use **Route 53 Failover Routing** with health checks.
- Replicate container images using **Amazon ECR replication**.
- Enable **S3 Cross-Region Replication** for critical data.
- Use **RDS Cross-Region Read Replicas** or an appropriate database replication strategy.
- Replicate secrets using **AWS Secrets Manager**.
- Perform **regular DR drills** (at least annually, or more frequently for critical systems).
- Monitor replication lag, health checks, and failover events with **CloudWatch** and **SNS**.
- Document recovery procedures and keep runbooks up to date.